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Worksheet – States of matter simulation

Grade 6 · Physics · Unit 1 Thinking about Energy · States of matter, kinetic energy of the particles, changes of state

Name: _____ Class: _____ Date: _____

school.taneski.com/science/g06/07-states-of-matter/

Open the simulation (scan the QR code), do the 4 parts with Marie and fill in this sheet as you go. Write temperatures with the unit °C.

Part 1 · Solid, liquid, gas (🧊 water: push the lid, pour it)

1 Copy the results table from the simulation.

State	Shape (🫙 pour)	Volume (🔧 push the lid)	Particles: how close?	Particles: how do they move?
Solid				
Liquid				
Gas				

2 Draw **9 particles** in each jar to show how they are arranged.

Solid (ice)	Liquid (water)	Gas (steam)

3 Why can a gas be squashed, but a liquid can't? Use the word **particles**.

Part 2 · Melting and freezing (🧊 ice · 🍫 chocolate · 🕯️ candle wax)

4 Copy the melting points from the results table.

Substance	Water (ice)	Chocolate	Candle wax
Melts at (°C)			

5 While I heated the ice, the ⚡ kinetic energy bar went _____. The particles vibrated _____ until they broke out of their _____: the ice _____.

6 Water freezes at _____. When a liquid freezes, its particles _____ energy, _____ down and get stuck in fixed places.

7 A piece of chocolate melts in your hand. Where does the energy come from? Explain.

Part 3 · Evaporating, boiling, condensing (💧 water)

8 Below 100 °C a few fast particles escaped from the _____ of the water. This is called _____. At _____ all the water _____ and the particles spread out to fill the whole jar.

9 Write **one** difference between evaporating and boiling.

10 After a hot shower the cold bathroom mirror is covered with tiny drops. Explain with **particles**. Which change of state is it?

Part 4 · Same rules, different substances (🔩 iron)

11 Complete the table.

Substance	Melts at	Boils at	State at 20 °C
Water			
Chocolate			
Candle wax			
Iron			

12 At 50 °C: chocolate is a _____, candle wax is a _____, iron is a _____.

13 Why is iron a solid at 20 °C, but water is a liquid?

Summary · changes of state

14 Write the name of each change of state on the arrows (top row: 🔥 heating · bottom row: ❄️ cooling).

SOLID	🔥 _____ → ← _____ ❄️	**LIQUID**	🔥 _____ → ← _____ ❄️	**GAS**

15 When we heat a substance, its particles get _____ kinetic energy and move _____. When we cool it, they _____.

16 One thing that surprised me in this simulation: